CBE423 Assignment 2

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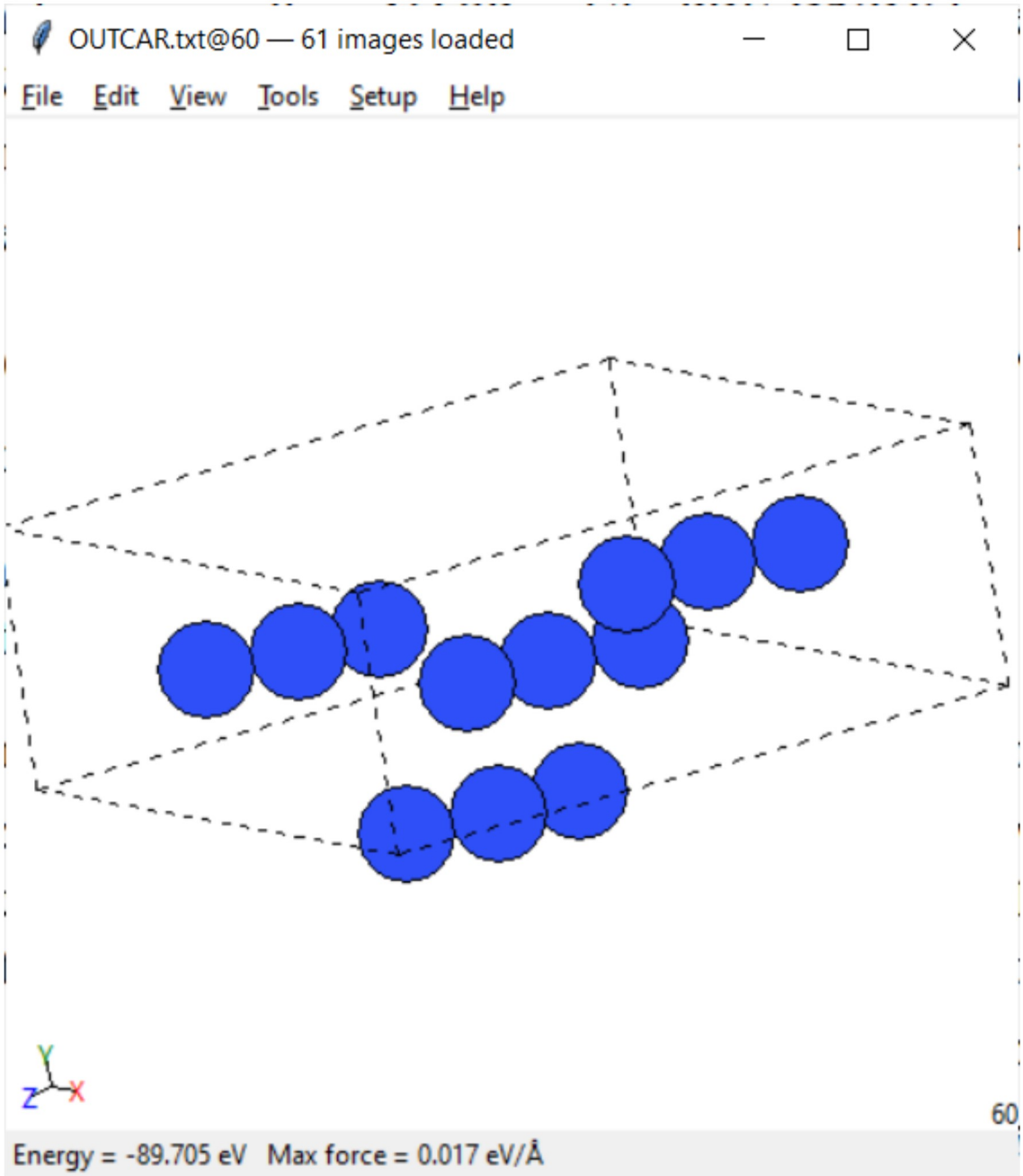
1

1.1

PBE is a GGA functional.

1.2



1.3

In [7]:
```
from ase.io import read

# load data
atoms = read('./problem_1/OUTCAR.txt',index='-1')

# get cell parameters
params = atoms.get_cell_lengths_and_angles()

print(params)
```

```
[  6.25365154   3.68162549   8.65560159  90.         117.29399588
  90.        ]
```

```
C:\Users\kevy0\AppData\Local\Temp\ipykernel_29280\3219290680.py:7: DeprecationWarnin
g: Please use atoms.cell.cellpar() instead
  params = atoms.get_cell_lengths_and_angles()
```

Based on `OUTCAR.txt` , $a = 6.254\ldots\mathrm{A}$, $b = 3.683\ldots\mathrm{A}$, $c = 8.656\ldots\mathrm{A}$, $\alpha = 90\,\mathrm{deg}$, $\beta = 117.294\,\mathrm{deg}$, and $\gamma = 90\,\mathrm{deg}$.

2

2.1

a) Charge: 0, Spin Multiplicity: 2 b) Charge: 0, Spin Multiplicity: 1 c) Charge: 0, Spin Multiplicity: 1

2.2

a) B3LYP is a hybrid functional. b) If def2-TZVP is too computationally intractable, a lower accuracy basis set such as def2-SVP can be used instead.

2.3

For $\cdot N_3$, the total energy is $-164.129Eh$ and the Gibbs free energy is $-164.136Eh$. For N_6 , the total energy is $-328.305Eh$ and the Gibbs free energy is $-328.309Eh$. For N_2, the total energy is $-109.521Eh$ and the Gibbs free energy is $-109.534Eh$.

2.4

In [ ]:
```python
from ase.io import read

atoms = read('./problem_2/n6.xyz')
```

```
Angle 5-0-1: 38.24°
Angle 0-1-2: 141.76°
Angle 1-2-3: 20.29°
Angle 2-3-4: 2.59°
Angle 3-4-5: 10.26°
Angle 4-5-0: 152.03°
```

Using `ase gui n6.xyz` and 'ase gui OUTCAR.txt`.

Method: ORCA	1	2	3	4	5	6
Distance from i+1 (Å)	1.128	1.238	1.440	1.238	1.128	0
Angle w.r.t neighbors (deg)	N/A	172.3	109.6	109.6	172.3	N/A

Method: Figure 4a	1	2	3	4	5	6
Distance from i+1 ($\mathrm{\AA}$)	1.138	1.251	1.460	1.251	1.138	0
Angle w.r.t neighbors (deg)	N/A	172.5	107	107	172.5	N/A

Method: Problem 1	1	2	3	4	5	6
Distance from i+1 ($\mathrm{\AA}$)	1.142	1.243	1.415	1.243	1.142	0
Angle w.r.t neighbors (deg)	N/A	167.3	111.9	107	172.5	N/A

Overall, the results using ORCA are similar to those of Figure 4a and Problem!

2.5

```
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VIBRATIONAL FREQUENCIES
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Scaling factor for frequencies =  1.000000000  (already applied!)

     0:       0.00 cm**-1
     1:       0.00 cm**-1
     2:       0.00 cm**-1
     3:       0.00 cm**-1
     4:       0.00 cm**-1
     5:       0.00 cm**-1
     6:   -1062.98 cm**-1  ***imaginary mode***
     7:      63.15 cm**-1
     8:     139.68 cm**-1
     9:     169.70 cm**-1
    10:     284.26 cm**-1
    11:     367.40 cm**-1
    12:     490.31 cm**-1
    13:     659.28 cm**-1
    14:     732.77 cm**-1
    15:    1179.28 cm**-1
    16:    2141.80 cm**-1
    17:    2163.68 cm**-1
```

There is exactly one vibrational mode of magnitude $1062.98\ \mathrm{cm}^{-1}$. This makes intuitive sense as a transition state/saddle point should have at least one imaginary frequency from a geometric perspective (at least one negative curvature). The magnitude is also reasonable N_6 is likely to decay to N_2; a large magnitude reflects this "attraction" to decay.

2.6

Note to convert Eh to $\frac{\mathrm{kcal}}{\mathrm{mol}}$, multiply by 627.509. Also, note that Figure 4a uses the Gibbs

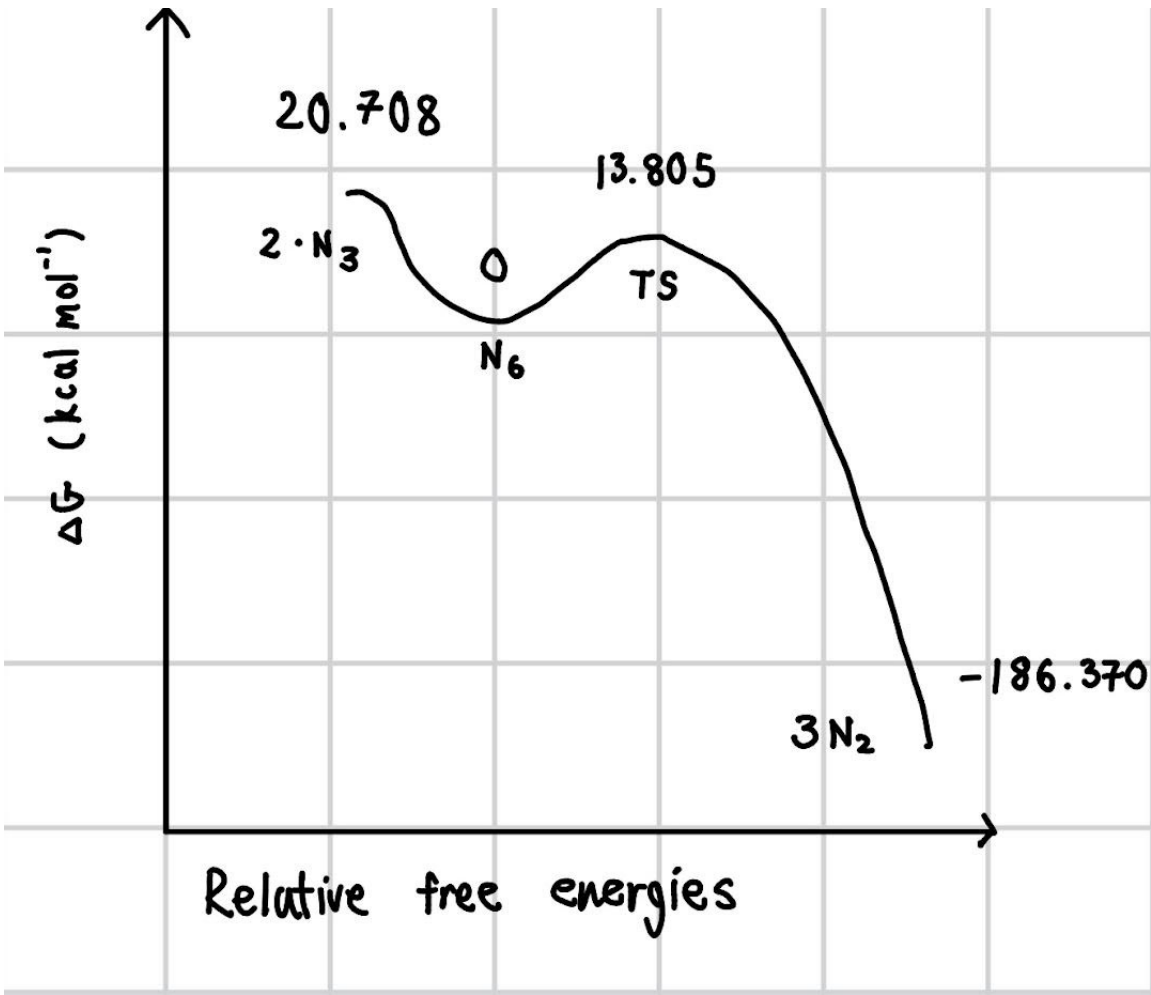
Free Energy of N_6 as a baseline.

Thus, the following formula is used:

$$\Delta G_{molecule} = (G_{molecule} - G_{N_6}) \cdot 627.509$$

$$\Delta G_{2 \cdot N_3} = (2 \cdot G_{N_3} - G_{N_6}) \cdot 627.509 = (2 \cdot -164.136 - (-328.305)) \cdot 627.509 = 20.708 \frac{\mathrm{kc}}{\mathrm{m}}$$

$$\Delta G_{N_6} = (G_{N_6} - G_{N_6}) \cdot 627.509 = 0 \frac{\mathrm{kcal}}{\mathrm{mol}}$$

$$\Delta G_{TS} = (G_{TS} - G_{N_6}) \cdot 627.509 = (-328.283 - (-328.305)) \cdot 627.509 = 13.805 \frac{\mathrm{kcal}}{\mathrm{mol}}$$

$$\Delta G_{3 \cdot N_2} = (3 \cdot G_{N_2} - G_{N_6}) \cdot 627.509 = (3 \cdot -109.534 - (-328.305)) \cdot 627.509 = -187.37($$